

Sofia Moreiras Pimazzoni

Unstalked: Digital Investigation service based on demand

SÃO PAULO
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Resumo

Moreiras Pimazzoni, Sofia. **Unstalked**. 2025. TCC (Graduação) – Curso Engenharia de Software, Instituto de Tecnologia e Liderança, São Paulo, 2025.

A Unstalked é uma startup de investigação digital que atende às diversas necessidades dos clientes, sendo estes tanto pessoas físicas como jurídicas. Este trabalho teve o objetivo de construir e maturar essa ideia, entendendo a situação da área nos dias atuais, definindo o modelo de negócios, modelos financeiros e estratégias de expansão, para que a startup pudesse entrar ao mercado no início de 2026. É importante mencionar que o entregável não é um produto, e sim um serviço em espécie de consultoria, no qual necessita de muito treinamento para ser realizado, e por este motivo também não é escalável. O início do desenvolvimento do projeto teve como foco estudar o mercado para analisar os concorrentes, definir os serviços, encontrar lacunas que se tornaram os diferenciais da startup e principalmente definir o modelo de negócio. A segunda parte da análise concentrou-se em três eixos de trabalho distintos. Primeiramente, realizou-se um estudo aprofundado dos processos internos, mapeando o fluxo completo da demanda à entrega. Em seguida, procedeu-se à criação do mockup para o site. Por fim, de forma independente, foi executado um caso de uso real que aplicou etapas detalhadas dos processos internos definidos. A terceira parte do trabalho teve um foco maior na definição dos modelos financeiros, estimativas de custo e receita, considerando projeções para os próximos cinco anos. Por último, de forma geral, a finalização do trabalho contou com planos de lançamento da startup ao mercado, KPIs e resultados esperados e finalização do desenvolvimento do site. Os resultados obtidos em todas as partes trouxeram muitos aprendizados, que envolveram principalmente a mudança de estratégias a fim de melhorar a apresentação do serviço e oportunidade ao cliente. Portanto, essas definições de conceitos base no processo de ideação da startup serviram de grande aprendizado para compreender suas lacunas atuais e como é possível agir para preencher esses espaços.

Palavras-Chave: investigação digital; OSINT; privacidade; discrição; sigilo.

Abstract

Moreiras Pimazzoni, Sofia. **Unstalked**. 2025. Final course project (Bachelor) – Course Software Engineering, Institute of Technology and Leadership, São Paulo, 2025.

Unstalked is a digital investigation startup that meets the diverse needs of both individual and corporate clients. The objective of this project was to develop and refine this idea, understanding the current situation in the field, defining the business model, financial models, and expansion strategies, so that the startup could enter the market in early 2026. It is important to mention that the deliverable is not a product, but rather a consulting service, which requires a lot of training to be carried out, and for this reason is also not scalable. The beginning of the project's development focused on studying the market to analyze competitors, define services, find gaps that became the startup's differentials, and, above all, define the business model. The second part of the analysis focused on three distinct areas of work. First, an in-depth study of internal processes was conducted, mapping the entire flow from demand to delivery. Next, a mockup for the website was created. Finally, independently, a real use case was executed that applied detailed steps from the defined internal processes. The third part of the work focused more on defining financial models, cost and revenue estimates, considering projections for the next five years. Lastly, in general, the completion of the work included plans to launch the startup on the market, KPIs and expected results, and completion of the website development. The results obtained in all parts brought many lessons, which mainly involved changing strategies in order to improve the presentation of the service and opportunity to the customer. Therefore, these definitions of basic concepts in the startup's ideation process served as a great learning experience to understand its current gaps and how it is possible to act to fill these spaces.

Keywords : digital investigation; OSINT; privacy; discretion; confidentiality.

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List of Abbreviations and Acronyms

1. OSINT - Open Source Intelligence;
2. IDEF0 - Function modeling methodology for describing manufacturing functions;

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1 Introduction

The field of digital investigation is an adaptation of the traditional detective's role, with the only difference being the work environment. While detectives work in the field, digital investigators only use their computers and cell phones with internet access to carry out their work.

In addition to that, today information and data are the world's most valuable asset, and people are increasingly going after it. That's why they pay people with knowledge of digital investigation to find what they can't on the internet.

1.1 Context and Motivation

The startup's area of expertise is Digital Forensic Investigation and Expertise, focusing on the acquisition, preservation, analysis, and presentation of digital evidence in a legally admissible manner.

Its work revolves around understanding the needs of its clients and seeking information that can help them solve problems, ranging from corporate fraud to investigations focused on understanding the level of exposure of individuals on the internet.

Thus, the customer's decision to hire our services is justified by our technical expertise and the way in which we use it to meet their personal needs and solve their problems.

With this in mind, we identified a market opportunity in the gap between the technical complexity of investigations and the time required to carry out such activities.

Finding hidden information on the internet requires in-depth technical knowledge (which most people and organizations do not have) and a lot of time and patience. This is where we come in: we are the high-performance technical resource that individuals need to obtain legally valid evidence and make informed decisions.

1.2 Problem Definition and Value Proposition

In the digital world, finding relevant and sensitive information can be a major challenge for individuals and organizations. Police, lawyers, companies and even individuals often need to locate specific data for investigations, legal proceedings, strategic decision-making or fraud protection. However, this information is not always easy to access, and many are unaware of where to look for it or whether it even exists.

In addition, today information and data are the world's most valuable asset, and because there is a huge amount available on the internet, searching for information becomes a complex and time-consuming task. Without adequate tools and specialized technical knowledge, the collection and analysis of this data can be inefficient, inaccurate and, in some cases, even illegal if done improperly.

Faced with this scenario, there is a growing demand for professionals specializing in digital investigations, capable of mapping information sources, cross-referencing data and carrying out precise analyses to support criminal investigations, legal disputes, corporate audits, fraud prevention and other specific cases. The startup is here to fill this gap, offering a structured, ethical and effective service to help those who need to access strategic information in a reliable and legal way.

At the end, customers will be able to identify and mitigate risks before they become more serious problems and will have a sense of control and security regarding their online exposure, which ultimately boils down to having a more private life.

1.3 Objectives of the Work

In general, the objective of this work is to develop the idea of a startup that offers digital investigation services, preparing all the necessary processes for its launch on the market in the first half of 2026.

To achieve the overall objective, specific objectives were defined. In the first part of the work, the focus was on ideation and market research, containing analyses such as Value Proposition Canvas, Personas, Risk Matrix, Blue Ocean Matrix, and Business Model Canvas, as well as market-oriented analyses, such as hypothesis creation, idea validation, and competitor research.

The second part of the development focused on three distinct areas of work. First, an in-depth study of internal processes was carried out, mapping the entire flow from demand to delivery using the IDEF0 diagram, and then detailing each specific process using BPMN modeling. Next, the wireframe and mockup for the website were created. Finally, a real use case was independently executed, applying detailed steps from the defined internal processes.

The third stage focused on financial models, i.e., it began with the definition of costs and business model, so that revenue and cash flow projections for the next five years could then be made. In addition to the financial deliverables, two important non-financial deliverables were completed. First, with the mockup of the website finalized, development of the landing page began. Second, a detailed document outlining business expansion strategies was prepared, since traditional methods such as social media creation and paid traffic will not be used.

Finally, the project was concluded with the planning of the company's launch on the market, involving the finalization of the landing page, creation of the sales pitch, launch plan and customer onboarding, and redefinition of the service billing method for a more attractive strategy for the customer.

1.4 Justification and Contributions

The project involves the conception and development of a startup that provides an on-demand service. This means that the company was created to solve specific and timely needs of our client.

The idea came from both partners' passion for investigation. Over time, we noticed that many close friends had common everyday problems and needed to find answers, but did not know how or where to look. These situations helped us come to the conclusion that we could help them solve such problems. Analysis of the sector revealed a marked lack of technical expertise, highlighting that digital investigation is a field operated by a restricted circle of specialists. Despite the shortage of professionals, their work plays a vital role in conducting complex research and supporting high-impact investigative processes.

In this sense, the desire arose to create a business that would combine passion with real needs, and thus Unstalked was born.

1.5 Work Structure

The following pages, totaling 50, will detail the project in Chapter 2 with the development of the solution, and finally conclude it in Chapter 3. The second chapter is divided into six parts and addresses the overall development of the solution, from initial hypotheses to validations and results.

- Section 2.1: covers the initial hypotheses, involving the problem, the solution, and the validation that one completes the other;
- Section 2.2: covers the analysis of market size and recognition of the startup's target audience;
- Section 2.3: covers the analysis of competitors, both direct and indirect;
- Section 2.4: covers more technical topics of the solution such as requirements, architecture, use cases, and others;
- Section 2.5: covers business aspects and market definitions, such as financial planning, marketing strategies, Business Model Canvas, and others;
- Section 2.6: covers project validation topics and future concerns such as KPIs and risks to be considered;

Chapter 3 then brings together the conclusion of the project, with general and specific observations on the main lessons learned during development.

Finally, the document ends with the project references and appendix sections.

2 Solution Development

Chapter two addresses the entire development of the project in detail, tracing the path that transforms an identified market gap into a viable and structured digital investigation solution. To this end, the section is divided into fundamental pillars that underpin the robustness of the business: initially, the premises and hypotheses that guide the value proposition and pricing model are established; then, an in-depth analysis of market size and the competitive landscape is carried out to define the target audience and strategic personas.

In addition, the architecture of the technological solution is detailed, based on the OSINT methodology and human analytical rigor, as well as the respective business plan, which translates the operation into financial viability and expansion guidelines. The chapter concludes with the validation and analysis of results, transparently presenting performance indicators, mitigated risks, and critical learnings obtained through direct contact with the market, consolidating the basis for the future implementation of the company.

2.1 Definition of Market Assumptions and Hypotheses

This subsection describes the market assumptions and business hypotheses that guided development, focusing on understanding the target audience's pain points and the value proposition offered.

2.1.1 Problem Hypothesis

Based on the problem presented above, it is assumed that *individuals and corporations want to discover hidden information on the internet for their personal use, but do not have the technical knowledge to do so, nor the time for such activity, so they are willing to pay professionals with the knowledge and time in exchange for the information.*

The depth of this pain lies in the critical and often confidential nature of the information the client seeks. In today's digital environment, crucial evidence for personal topics (such as family issues, divorces, etc.) or business topics (such as suspected fraud, executive investigations, etc.) is often buried in layers of data, hidden in image and document details, or scattered across devices that are difficult to access. The problem is not only the difficulty of finding the information, but also the

risk of compromising the legal validity of the evidence during the collection attempt. Without adequate forensic technical knowledge, the information obtained may be considered inadmissible or contaminated in a court case, rendering the effort and search useless. Therefore, the lack of expertise creates paralysis in resolving a high-impact personal or business problem.

2.1.2 Solution Hypothesis

Considering the hypothesis for the problem presented, it is assumed that *individuals and corporations are willing to invest in digital investigation services on the premise that the application of OSINT methodology and specialized tools by professionals with high knowledge in the area represents the fastest, most accurate, and legally defensible way to extract and validate the hidden information necessary to solve their problems.*

The superiority of the proposed solution lies in the strategic application of forensic technical knowledge and the full-time dedication to the investigation. The OSINT methodology, although based on open sources, requires the human ability to reason, correlate, and pivot layered information (a skill that automated software does not possess).

2.1.3 Value Hypothesis

Considering the problem and the solution, it can be assumed that the *customer's willingness to pay for digital investigation services is directly proportional to the complexity of solving the problem and the perceived value of the information obtained.*

In other words, the more valuable and difficult to extract the answer is, the greater the acceptance of the value proposed by the service. In addition to intrinsic complexity, crucial intangible factors such as strict confidentiality, professional credibility, investigator experience, and clarity in the presentation of results act as value multipliers, reinforcing the price justification and significantly increasing the chance of closing the deal.

The client's investment is justified, as they are acquiring a meticulous and exhaustive investigation process, where the professional uses their experience to identify non-obvious search paths, perform social engineering on profiles, and apply contextual filters that ensure the information is relevant and reliable. In addition, the professional is trained to document the chain of custody of evidence manually and rigorously, ensuring that each piece of information collected is legally defensible,

which eliminates the risk of contamination and makes the client's effort the most strategic and reliable path to solving their critical problems.

2.2 Market Sizing and Analysis

Understanding the scale of the market is essential to validate the commercial viability and growth potential of this initiative. This subsection presents market size estimates based on an analysis of the current information security landscape and the growing demand for specialized digital intelligence. In addition to quantitative projections, a central section is devoted to segmentation and the ideal customer profile, detailing the personas that make up the target audience.

2.2.1 Market Size Estimations

As mentioned in the document summary, this project does not involve the creation of a scalable product, but rather the provision of a service, and therefore the size of the market will depend on demand and contract closures.

An important topic to consider is that the complexity of demand impacts market size, since a month, for example, can be filled with three small projects, but also with one large project, and this does not necessarily mean an increase or decrease in market size.

2.2.2 Customer Segmentation and Profiling

The target customer segment consists of individuals belonging to classes A and B, mostly aged between 30 and 50, located in large urban centers. People over 18 (outside the above age range) may also be served, but they are not the focus and require greater caution in analyzing demand. This group is in a context where they are more exposed to the need for the service, such as involvement in legal cases, the need to live a more private life, etc., which makes them willing to pay for a specialized technical service that guarantees the speed and legal confidentiality of digital evidence.

Based on the segment discussed above, two personas were created, Mariana Garcia and Gabriel Paiva:

Mariana Garcia

 MARIANA GARCIA EXECUTIVE Age: 42 Job title: Chief Financial Officer Education: economics and administration	BIOGRAPHY Mariana Garcia is 42 years old and CFO of a large company in the financial sector. With an established career, she is frequently invited to talks, interviews and industry events, which makes her a public figure in the corporate world. Mariana is highly connected, using social networks for professional networking and to keep up with market trends. However, until recently, she had never been deeply concerned about her digital security.	CONTEXT Recently, Mariana had one of her social networks hacked, which sparked a warning about how much of her personal information is exposed on the internet. Because she holds a high-visibility position, she realized that she could be the target of targeted attacks, such as scams, extortion and even personal threats. This made her uneasy, as she had never stopped to think about the amount of information available about her online and how it could be used against her.
	PAINS • Concern about being the target of scams or extortion attempts due to her high-level position. • Fear that personal information, such as address or routine, will be used against them. • Feeling a lack of control over what is already exposed on the internet. • Difficulty finding a reliable service to help her understand and reduce her digital exposure. • Anxiety at the thought that malicious people may have access to sensitive data about her without her knowing.	MOTIVATIONS • Have more control over your privacy and reduce your digital exposure. • Know exactly what personal information is accessible and how to minimize it. • Feel safer when using social networks and other digital platforms. • Avoid risky situations such as fraud, harassment or threats. • Share this solution with other executives and friends who may be in the same situation.

Image 1: persona Mariana Garcia

Biography

Mariana Garcia is 42 years old and CFO of a large company in the financial sector. With an established career, she is frequently invited to talks, interviews and industry events, which makes her a public figure in the corporate world. Mariana is highly connected, using social networks for professional networking and to keep up with market trends. However, until recently, she had never been deeply concerned about her digital security.

Context

Recently, Mariana had one of her social networks hacked, which sparked a warning about how much of her personal information is exposed on the internet. Because she holds a high-visibility position, she realized that she could be the target of targeted attacks, such as scams, extortion and even personal threats. This made her uneasy, as she had never stopped to think about the amount of information available about her online and how it could be used against her.

Pains

- Concern about being the target of scams or extortion attempts due to her high-level position.
- Fear that personal information, such as address or routine, will be used against them.
- Feeling a lack of control over what is already exposed on the internet.
- Difficulty finding a reliable service to help her understand and reduce her digital exposure.
- Anxiety at the thought that malicious people may have access to sensitive data about her without her knowing.

Motivations

- Have more control over your privacy and reduce your digital exposure.
- Know exactly what personal information is accessible and how to minimize it.
- Feel safer when using social networks and other digital platforms.
- Avoid risky situations such as fraud, harassment or threats.
- Share this solution with other executives and friends who may be in the same situation.

Gabriel Paiva

 GABRIEL PAIVA ADVOGADO Age: 34 Job title: criminal lawyer Education: law with a specialization in criminal and civil law	BIOGRAPHY Gabriel Paiva, 34, is a lawyer specializing in criminal and civil law. With over 12 years of experience in the field, he has built a solid career defending clients in complex cases. He works in a renowned law firm, often dealing with cases that require in-depth investigation to gather evidence and strengthen the defense or prosecution. He is meticulous, strategic and always looking for new tools to improve his work.	CONTEXT Gabriel is currently working on a court case that could define his client's future. The defense depends on obtaining a crucial piece of evidence, but traditional forms of investigation have not brought results. He has tried various approaches - database queries, traditional forensics and testimony - but feels that there is hidden information that could change the course of the case.
PAINS • Difficulty in finding relevant evidence to strengthen your argument in court. • Have exhausted all conventional search resources and face difficulties identifying new sources of relevant information. • Bureaucracy and legal limitations make obtaining certain information very slow. • Needs a reliable and confidential service that respects legal standards.	MOTIVATIONS • Looking for a more efficient and strategic way to obtain evidence that can strengthen your case and ensure a solid argument in court. • Find evidence that can be used legally to defend your clients. • Work with experts who understand the weight of confidentiality in legal proceedings. • Resolve cases more quickly and effectively, gaining recognition in the field.	

Image 2: persona Gabriel Paiva

Biography

Gabriel Paiva, 34, is a lawyer specializing in criminal and civil law. With over 12 years of experience in the field, he has built a solid career defending clients in complex cases. He works in a renowned law firm, often dealing with cases that require in-depth investigation to gather evidence and strengthen the defense or prosecution. He is meticulous, strategic and always looking for new tools to improve his work.

Context

Gabriel is currently working on a court case that could define his client's future. The defense depends on obtaining a crucial piece of evidence, but traditional forms of investigation have not brought results. He has tried various approaches - database queries, traditional forensics and testimony - but feels that there is hidden information that could change the course of the case.

Pains

- Difficulty in finding relevant evidence to strengthen your argument in court.
- Have exhausted all conventional search resources and face difficulties identifying new sources of relevant information.
- Bureaucracy and legal limitations make obtaining certain information very slow.
- Needs a reliable and confidential service that respects legal standards.

Motivations

- Looking for a more efficient and strategic way to obtain evidence that can strengthen your case and ensure a solid argument in court.
- Find evidence that can be used legally to defend your clients.
- Work with experts who understand the weight of confidentiality in legal proceedings.
- Resolve cases more quickly and effectively, gaining recognition in the field.

Companies performance

The company's performance is validated by two central personas that encapsulate market demands:

1. The executive: this persona seeks the service as a form of risk management and reputational security. Her pain point is digital exposure and concern about the fragility of her online image. She invests to identify her level of exposure on the internet and how to make her life more private, requiring a service based on strict confidentiality and preventive analysis.

2. The lawyer: this profile seeks digital investigation as a strategic tool to win court cases. Their pain point is the lack of admissible evidence to support their client's case. They value technical methodology (OSINT) and an impeccable chain of custody, as the success of the service is directly linked to the acceptance of evidence by the judicial system.

In summary, our target customer segment is characterized by a willingness to pay for the reduction of legal risk and the transformation of uncertainty into actionable evidence, ensuring that the information obtained is the key to solving their critical problems, whether personal or professional.

2.3 Competitive Analysis and Differentials

To consolidate a strategic position, it is essential to analyze the ecosystem of direct and indirect competitors operating in the field of digital intelligence. This analysis assesses how existing solutions operate.

During the research, only a few direct competitors were identified, i.e. companies or agencies that offer exactly the same service as ours, or something very similar. However, there are a large number of private detective agencies that, although they have a certain similarity, don't fully fit the scope of our proposal.

To simplify the comparison, I've drawn up a table that highlights which competitors require the most attention, as well as analyzing their strengths, weaknesses and other relevant aspects.

It's worth noting that this research has limitations because, as this is a service that involves security, companies in the sector tend to operate in a more secretive manner, making it difficult to obtain details about their operations and methodologies. Given this scenario, the analysis of competition was based mainly on the way these companies present themselves to customers through their websites.

Nome	O que oferece?	tipo de concorrente	Tempo no mercado	Pontos fortes	Pontos fracos	site
Bruno Figueira	It offers digital investigation services, mainly OSINT of all kinds, both low complexity, such as finding addresses, telephone numbers, etc. of an individual, and high complexity, such as making connections between people, finding criminals, etc.	direct	doesn't mention	-He was one of the pioneers in the digital research market; -He has created several courses on the subject, which shows his mastery; -He is well known in this market because he has been involved in cases involving famous people, which increases confidence in his service;	-It has built up a negative image over time with its course platform, because it promised and didn't deliver, as well as many other issues that can lead customers to no longer trust its services; -It no longer has a website or any other means of communicating its service, but only a YouTube channel, which it publishes weekly;	it doesn't have
Antônio Miranda	He offers a detective service, i.e. he uses the physical world to find out information, whether for family, business or private cases, among others. He carries out investigations if necessary, but always guarantees that the contractor will not be exposed and will have maximum security. His aim is to help the contractor obtain evidence to be used in court.	indirect	over 20 years' experience in private investigation	-Many years of experience in the market, which gives clients confidence; -Organized website with all the services it provides, covering various possibilities;	-Marketing for hiring a bit forced; -The links to your social networks on the website don't work;	https://mirandadetective.com.br/index
Detective conjugal internacional	Promises to find out everything the contractor wants to know about their spouse	indirect	doesn't mention	-They serve several countries	-Very small site with very little information, which does not give confidence in the service; -From the structure of the site, it looks like a scam	https://detectivedessocios.com.br/
Agência detetive Rodrigues	Because it is an agency and therefore has more manpower, Detective Rodrigues covers the following services: marital investigations, family investigations, abuse investigations, business investigations, counter-espionage investigations and the sale and rental of devices. It is important to note that all investigations are carried out in the field.	indirect	over 20 years' experience in investigations	-Very well-structured website with several explanations and demonstrations of the service, which gives customers confidence; -She has a blog, where she publishes things related to her service, which complements the information on her website and shows that she has mastery of the subject; -Has very well-defined services, and puts a lot of emphasis on marital cases, as these are the ones that attract the most clients -He makes it clear why the client should hire his service -Certified by the National Association of Detectives and Private Investigators of Brazil -A pioneer in digital forensics -Recommended by JustBrasil in an article on digital forensics	-The About Us page doesn't work, but this hasn't harmed the company's credibility too much, as it goes over this point lightly on the home page	https://www.detectivedesrodigos.com.br/index.html
Cyber Experts	They offer services in digital and computer forensics, data protection and auditing, online reputation and privacy, forensics and cryptocurrency trading	direct	more than 20 years of legal expertise and technical assistance in computers and information technology	-They have an academy area with courses and training -They have a blog, which publishes things related to their service, which complements the information on their website and shows that they have mastery of the subject; -They have a page dedicated to explaining what digital forensics is, as it is not very well known; -They are serious in their words and in their presentation of the company, which gives the client more confidence; -They use the name digital forensics instead of digital investigation, which makes the service sound more serious;	Site with a lot of text, which makes reading tiring and doesn't highlight the most important information	https://cyberexperts.com.br/
Sabre inteligência e investigação	It has a strong focus on corporate investigations, offering a range of services in this area, such as fraud, employee, executive, supplier, asset and background checks investigations. It also offers computer forensics and cell phone forensics. In general, the service this company provides is related to uncovering crimes and fraud within the corporate world.	direct	more than 20 years	-Present the company's values and mission	-The visual identity of the site is not in keeping with the service, as the colors and images used are very reminiscent of a corporate health service, rather than a research service -The information provided on the site is not well presented, with a relative amount of text that is very similar -The blog page has no content, and once accessed it disfigures the entire site	https://sabreinteligencia.com.br/sobre-nos/area-de-atuacao
STW Brasil	They offer digital forensics services, which are more investigative, educational, offering training in the area through the Digital Forensics Academy, and auditing, consultancy, cybersecurity and support, which are more focused on the business sector.	direct	20 years of excellence in technology and security	-They have a blog, where they publish things related to their service, which complements the information on their website and shows that they have mastery of the subject; -They use the name digital forensics instead of digital investigation, which makes the service sound more serious; -They have several certifications that prove the quality of the service and the professionals -They have an educational area (Digital Forensics Academy), specialized in training individuals who want to work in the area -They have a space on their website dedicated to showing clients who have already hired the company and institutions trained by their academy -They have a podcast channel -They use the name digital forensics instead of digital investigation, which makes the service sound more serious;	-It exposes 4 of the professionals who work in the company with a high position, and since it is a company that provides security services, it is not recommended, since the data is exposed and this can damage the company's image in the eyes of the client.	https://stwbrasil.com/

Table 1: competitors analysis

2.3.1 Analysis of indirect competitors

With regard to indirect competitors, the main difference comes in the approach used to gather information. While my service is based on OSINT (Open Source Intelligence) techniques, exploiting open sources on the internet to gather and relate evidence, indirect competitors work directly in the field. They travel to strategic locations to observe people, capture images and collect evidence in person, making it a more physical investigation-oriented alternative.

Considering this, and that the analysis was based on the website of these companies, there are some positive points that I think could be added to my website, and they would be:

- Organize and explain very well each case investigation we accept;
- Many competitors, including direct ones, maintain blogs, which is an interesting strategy, as it demonstrates expertise in the subject and strengthens customer confidence;
- Making it clear why the client should hire my service;

2.3.2 Analysis of direct competitors

Direct competitors, on the other hand, mostly offer additional services beyond digital investigation, working in different areas. This can be a competitive differentiator, both positive and negative, depending on the client's perception. While diversification can convey versatility and greater experience in the market, it can also create the impression that the company is not highly specialized in any one area. On the other

hand, a business focused exclusively on digital research can be seen as more specialized, but can also raise doubts about its experience and ability to meet varied demands.

Another relevant factor is that most of my direct competitors have been operating in the market for over 20 years and have accumulated extensive experience, which may represent a challenge for my entry into this area. In addition, they often refer to digital investigation as “digital forensics”, which I believe conveys a more professional and trustworthy image to clients, so it's worth considering approaching the subject in this way on the website.

A lot of positive points have been mapped out just by each company's website, and we are going to do a study on how to address them on our page as well, and they are:

- Again, building a blog also comes up frequently with direct competitors, so it will definitely be addressed;
- A page dedicated to explaining what the service is can be an effective strategy for attracting more customers. As mentioned earlier, many people don't know or fully understand what digital research is and how it can help them. By providing clear and accessible information, it is possible to increase interest and trust in the service, broadening the audience's reach;
- Certifications proving our level of experience;
- Company values and mission statement;

In addition to these points, two of my direct competitors offer an educational area, with courses and certifications in the field. I consider this to be a relevant differentiator because, as well as increasing the credibility of the service in the eyes of clients, it also provides an opportunity for those wishing to enter the sector to acquire knowledge, and this initiative could even represent an additional source of revenue. On the other hand, this isn't a viable avenue for me to explore at the moment, as I'm just starting out in the field and not only do I not yet have enough experience to teach other people, but I also don't have the manpower to structure this type of offer.

Finally, considering all this analysis, we found an opportunity to differentiate ourselves in the digital security resources used and in the specialization of confidential investigation processes within the context of the digital world. For reasons of competitive strategy and protection of know-how, the startup's explicit differentials cannot be disclosed at this time. Confidentiality is essential, as the startup has not yet been launched on the market and premature exposure of these elements could compromise its competitive advantage at the time of its official introduction.

2.4 Technological Solution

After conducting market research and understanding the startup's positioning and opportunities, the technical part of the project begins. It is time to develop the idea and then put it into practice and perform the necessary tests. This section details the necessary requirements and how they fit into the service architecture to guide the processes later defined in the IDEF0 diagram.

2.4.1 Requirements and Specifications

The functional and non-functional requirements of the investigation process are focused on ensuring quality and security during the execution of the processes.

One of the pillars that drives the startup is confidentiality, since during investigations, we have access to very personal information about our clients' lives, and it is extremely important to ensure that the software and tools we use keep the data stored securely. In this sense, the communication channels used must also meet this non-functional requirement.

Complementary to confidentiality, a non-functional requirement of access control is applied, since the data from an investigation is only accessible to the client and the specialists working on the case.

Another requirement is traceability, as it is necessary to organize the evidence in a timeline to keep the data organized and available for consultation whenever necessary during the investigation period.

Furthermore, non-repudiation is another non-functional requirement, since all documents and evidence must be legitimate and authentic in order to be legally viable and for customers to be able to use them as desired.

With regard to the landing page, the interface must meet usability and communicability requirements, ensuring that the presentation of services is clear and free of ambiguity. This includes the use of language appropriate to the target audience, legible typography, and a visual layout that highlights the main benefits, allowing for quick and efficient reading.

This is an important requirement, as many individuals do not understand what a digital investigation service is and its benefits, and a clear presentation increases the chances of achieving this goal.

2.4.2 Architecture and Technology

As this is a service, the architecture and technologies used revolve around which tools, platforms, and channels will be needed in each process, from the arrival of the request to the final delivery.

Throughout all communication with the customer, we will use applications similar to WhatsApp, Gmail, Outlook, and others, but which are proven to be private and secure in the market and work with communication confidentiality, so that there is no risk of compromising the integrity of either party.

With regard to document production, we will use tools similar to the Office suite and Google Docs and Sheets, but with the same concern for private and secure technologies as the media.

The technologies mentioned above are not open source and operate on a subscription basis, with the manufacturers being owned by the organizations themselves.

To maintain data traceability, we will use a tool called Hunch.ly, widely used by researchers in the field to preserve evidence found and keep it organized.

2.4.3 Development and Implementation (MVP)

To describe in detail how the investigation process will work within the startup, the IDEF0 diagram was developed, showing how we integrate the best tools and practices to act on the processes. In addition, to complement the MVP, a landing page was created, which will be one of the ways the startup will attract customers and create demand.

IDEF0 Diagram

To design IDEF0 modeling, it is necessary to divide the overall process into 4 parts: inputs, outputs, rules and controls, and process support mechanisms. In the context of my project, these parts applied as follows:

Input records:

- Demand request
- Meeting to get to know the client and understand the context of the investigation
- Contracting
- Client says how soon they want the investigation updates
- Demand for closure
- Internal demand for quality assessment
- Periodic updating of the portfolio

Output records:

- Records of requests (traceability)
- Minutes of the meeting for record
- Execution planning
- Internal kickoff
- Periodic report according to the date agreed with the client
- Final report with the result of the investigation
- Internal quality report
- Portfolio of cases

Records of rules and controls:

- I can't share confidential information about investigations with anyone other than the client
- I can't tell clients the means I use for investigations
- It is necessary to periodically update the client on the progress of the investigation
- The use of secure means of communication is indispensable

Records/process support mechanisms:

- Tools used for the investigation itself
- Tools used to record information during the investigation
- Important contacts within certain organizations
- Private tools for contacting the client

After defining these four points and allocating each one on the diagram, the boxes (rectangles) can be defined. They are nothing more than the sub-processes of the company's main objective, which in my case is the digital investigation service. In the way I have allocated the information above, I have 8 sub-processes:

- Prospect investigation service
- Detail scope of investigation
- Prepare work tools and strategies
- Start execution of service
- Monitor execution
- Close execution of service
- Evaluate quality
- Promote dissemination

These will be detailed in the BPMN modeling for a better understanding. Putting all this information together, you can assemble the IDFE0 diagram below. Its structure is made up of horizontal lines, with the input on the left pointing to the output on the right, vertical lines representing the rules, rectangles, which are the sub-processes of the digital investigation, and colored arrows, which indicate the dependence of

sub-processes, i.e. for one of them to be able to start, the previous one must be finished.

Below it is possible to understand how this structure works, but to get a quality view of the modeling, access the APPENDIX B on the Appendices section.

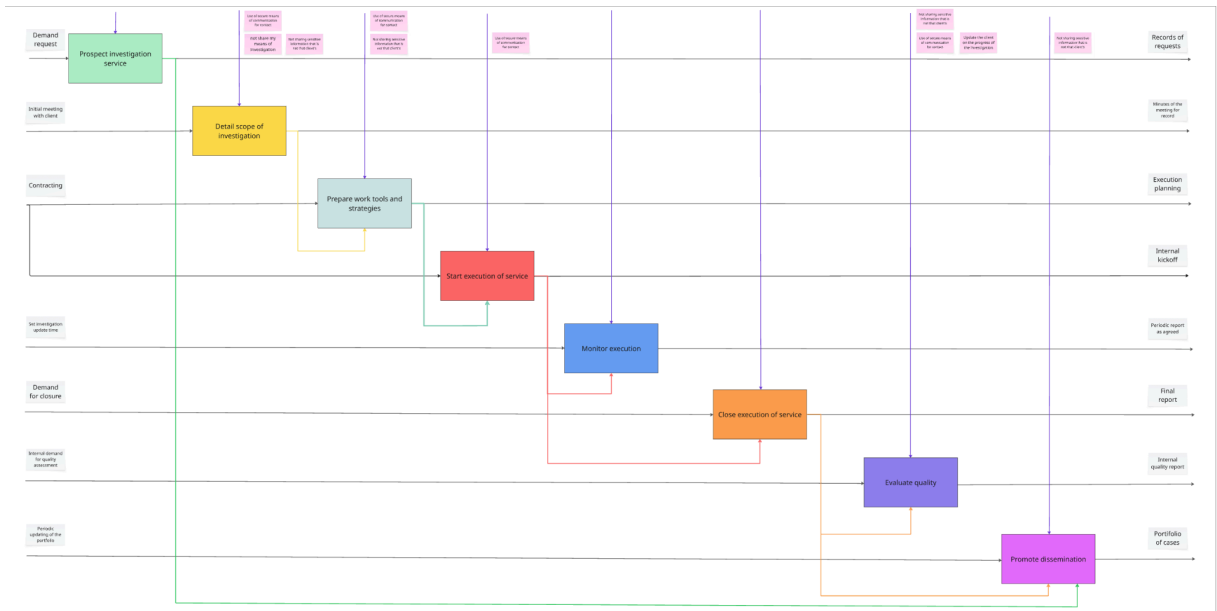


Image 3: IDEF0 processes diagram

Finally, it's important to note that there are two outcomes of the contract that may or may not be generated by the clients, these being the evaluation of the service and the recommendation to a friend. If, at the end of the contract, the client evaluates our performance and eventually recommends us to a friend who needs help, we would know where to improve and whether we are on the right track.

Landing Page

The development of a landing page is important to bring the business to life and serve as a gateway to attract customers at the beginning of the business, since the company will not have social media accounts. Therefore, great care was taken throughout its development to ensure that it reflected the company's image, and for this reason, a total of two versions were created. The first version was faithful to the mockup, but at the end of development, I realized that it lacked personality and contained too much text, which was tiring for the user. In this same version, some changes were made to spread the text out so that it didn't seem so overwhelming, but I wasn't very successful, and the website still looked more like a portfolio than a service sales page.

So, considering that there was no time to make a new mockup, I looked for inspiration that matched the personality I wanted and developed the new version based on that. The main changes revolve around the same fact: how to use elements that complement the text so that it is not tiring for the user.

For reasons of competitive strategy and protection of know-how, images from the website or its access link for comparing before and after changes cannot be disclosed at this time. Confidentiality is essential, as the company has not yet been launched on the market and premature exposure of these elements could compromise its competitive advantage at the time of its official introduction.

Even though the amount of text is the same in both cases (before and after the changes), simply rearranging the location of the elements gave the site a different and less tiring feel.

In the end, I managed to come up with a version with much more personality that sounded like a service sales website rather than a presentation website.

2.4.4 Testing and Technical Evaluation

During the second part of the project's development, we had the opportunity to test the methodology created in the IDEF0 diagram in a case study, to understand whether what had been defined fit reality and what points could be improved. Although it was called an "example", it turned out to be very real, since the request came from a real need of a friend.

The client hired us to help him with a lawsuit in which we needed to gather evidence to help him prove that an individual didn't need what he was claiming he was entitled to.

As it was an ongoing legal case, time was very short, so the investigation lasted a week, starting on 06/02/2025 and ending on 06/09/2025.

It is important to emphasize that no confidential information about the client or the case in general will be presented, and what needs to be presented will be masked. Therefore, the investigation processes and observations on what was done will be presented.

The request came through a conversation between my partner and a friend, in which he mentioned that he was starting out in the digital investigation field and the friend coincidentally had the need mentioned above. As a result, the meeting to detail the scope ended up taking place in person. After that, my partner accepted the proposal, since the initial objective seemed realistic, and we could work a little harder than planned to be able to deliver on time.

Considering this, the first observation I have to make is that the preparation of the tools and work strategies, in order to ultimately generate an execution plan, did not take place, due to the short timeframe. This is a very important consideration to make, because we don't know how much this has hurt us, since perhaps with good planning we could have found the information more easily, but it could also have taken up much-needed time for a result that wasn't as productive.

When the proposal arrived, the company and the way in which the proposals were to be carried out were not yet 100% structured, and so it was decided to charge the client a monetary amount if the investigation was successful, because in this way the client unconsciously has a positive image of us, since they only pay us if we get what they need, otherwise we work “for free”.

Bearing in mind that our strategy for attracting leads is word of mouth, starting with this “contract” model is a good idea. In the example case, it was agreed that the client would pay us if any of the information found was relevant to his legal case. Frankly, I don't think we managed to extract the maximum amount of information (which is understandable given our level of experience and the short time we had), and so it's likely that we won't be able to help our client with his objective. In this case, we won't receive anything, even though we've worked hard. In the future, we don't intend to continue with this strategy, as not finding anything doesn't mean we're bad investigators, sometimes the target just knows how to hide well.

Besides, we didn't have an official contract with the client, everything was agreed in the initial conversation and in other exchanges of messages. Considering that the client is a close friend, this situation is not such a problem, since there is trust in the relationship, but this can never happen with strangers. Even so, in the future, even with close friends, it is ideal to draw up a contract to ensure that both sides are protected.

The investigation did not go as expected. It would be unfair to say that we found nothing; a better description would be that we did not find enough. For reasons of client privacy, we cannot provide many details, but we did find evidence that helped him, although it was not enough to prove his case in court.

With regard to the periodic follow-up, as it was a very short period (one week), we didn't send the client any update reports, we just exchanged messages, on our part citing perhaps relevant findings and on his part complementing the case with new data.

Finally, we generated the final report for the client and delivered it. I think the biggest mistake in this process was that I didn't review the report to be delivered and the information we found in general with my partner. With the short deadline and my other priorities, I didn't have time to work on it with him, so we were out of alignment

on this point, which was very damaging, since after sending it to the client, we went to review and it was far below our quality standards.

With the pressure and the short deadline, we preferred to investigate until the last second and send the client a report made by the software we use, rather than produce a document manually. Realizing this after it had been delivered worried us, because the report made by the software doesn't present the information clearly, and even though we're not being paid, it was an opportunity to prove that our work has quality from start to finish.

Finally, there was internal discussion about the investigation process, to assess its strengths and weaknesses and propose improvements.

In addition to the actual use case, to conduct a demonstration with potential customers, we observed a post made in reels mode by an influencer with over twenty million followers on Instagram. The video in question showed areas near his home, such as the common area under the building and nearby establishments.

With that, we figured it would be possible to find out where this place was, and with a simple Google image search, we found exactly where the influencer lives. This is very concerning, because as ethical people, we have no negative influence on this data, but malicious individuals can track patterns and monitor behavior to take unethical action. This was a good case to prove how important our work is, especially in today's increasingly digital world.

2.5 The Business Plan

The Business Plan summarizes the company's strategic vision, integrating market analysis, technical operations, and financial objectives into an executive roadmap. This section serves as a feasibility map for the project, describing how the cost structure, revenue sources, and expansion strategies connect to ensure the company's sustainability and growth in the specialized research sector.

2.5.1 Market and Competitor Analysis

As described in more detail in section 2.2.2, the target market for the service consists of a select audience from classes A and B, mostly between the ages of 30 and 50, who live in large urban centers and face situations of high exposure, such as legal proceedings, the need for extreme privacy, or the resolution of personal problems based on digital evidence. This customer profile values agility and legal certainty, demonstrating a willingness to invest in specialized technical solutions that guarantee absolute confidentiality and the validity of digital evidence.

To meet these needs, the business is based on the OSINT (Open Source Intelligence) methodology, transforming scattered information into concrete and admissible evidence. While executive Mariana (persona) seeks control over her digital exposure to avoid extortion, lawyer Gabriel (persona) seeks to overcome bureaucratic limitations through meticulous technical investigation. Both converge on the need for a service that offers not only data, but security and predictability in high-pressure contexts, ensuring that the information obtained is the key to solving critical problems.

With regard to competitor analysis, more details can be found in section 2.3, but in general, there is a clear distinction between traditional investigators, who focus on physical and field activities, and established digital forensics companies. Although the longevity of direct competitors (many with more than 20 years in the market) represents an initial challenge, there is a significant opportunity in hyper-specialization. The adoption of more professional terms, such as “computer forensics,” and an exclusive focus on the digital environment allow the startup to position itself as a superior technical authority, differentiating itself from generalist agencies that may seem less focused on the digital investigation niche because they also have other fronts.

2.5.2 Business Model (Business Model Canvas - BMC)

This section presents the strategic structure of our operation, designed to convert digital intelligence and OSINT techniques into real value for the target audience. The model details how we integrate forensic rigor with reputational protection, ensuring an agile, confidential, private business structure focused on transforming complex data into answers and evidence for our clients.

For a more detailed view of the image below, see APPENDIX C in the Appendices section.

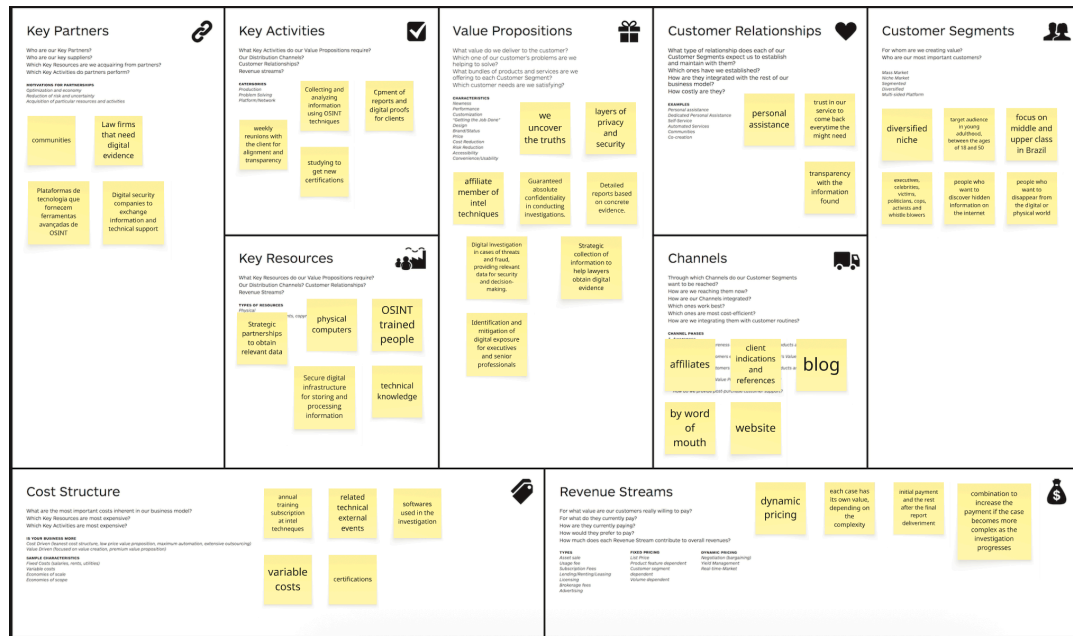


Image 4: Business Model Canva

Key partners

- Communities;
- Law firms that need digital evidence;
- Plataformas de tecnologia que fornecem ferramentas avançadas de OSINT;
- Digital security companies to exchange information and technical support;

Key Activities

- Collecting and analyzing information using OSINT techniques;
- Content of reports and digital proofs for clients;
- Weekly reunions with the client for alignment and transparency;
- Studying to get new certifications;

Key resources

- Strategic partnerships to obtain relevant data;
- Physical computers;
- OSINT trained people;
- Secure digital infrastructure for storing and processing information;
- Technical knowledge;

Cost Structure

- Annual training subscription at intel techniques;
- Related technical external events;
- Softwares used in the investigation;
- Variable costs;

- Certifications;

Value Propositions

- We uncover the truths;
- Layers of privacy and security;
- Affiliate member of intel techniques;
- Guaranteed absolute confidentiality in conducting investigations;
- Detailed reports based on concrete evidence;
- Digital investigation in cases of threats and fraud, providing relevant data for security and decision-making;
- Strategic collection of information to help lawyers obtain digital evidence;
- Identification and mitigation of digital exposure for executives and senior professionals;

Customer Relationships

- Personal assistance;
- Trust in our service to come back every time they might need;
- Transparency with the information found;

Channels

- Affiliates;
- Client indications and references;
- Blog;
- By word of mouth;
- Website;

Customer Segments

- Diversified niche;
- Large audience in young adulthood, between the ages of 18 and 50;
- Focus on middle and upper class in Brazil;
- Executives, celebrities, victims, politicians, cops, activists and whistle blowers;
- People who want to discover hidden information on the internet;
- People who want to disappear from the digital and physical world;

Revenue Streams

- Dynamic pricing;
- Each case has its own value, depending on the complexity;
- Initial payment, and the rest after the final report deliverment;
- Combination to increase the payment if the case becomes more complex as the investigation progresses;

2.5.3 Marketing and Sales Strategy

In the current scenario, market expansion depends not only on large investments in marketing, but also on building trust, credibility, and genuine connections. For businesses that value discretion and ethics, creating strategies based on solid relationships and the dissemination of knowledge proves to be more effective and sustainable than traditional advertising campaigns.

With that in mind, the main launch strategy will be to go live with the landing page and start implementing the defined customer acquisition strategies.

As our main strategy for promoting our service, we are committed to always delivering quality service in order to become a benchmark in our customers' network. This will allow them to refer friends who may need our service, and a referral from someone they trust is one of the best ways to get a customer to the end of the sales funnel.

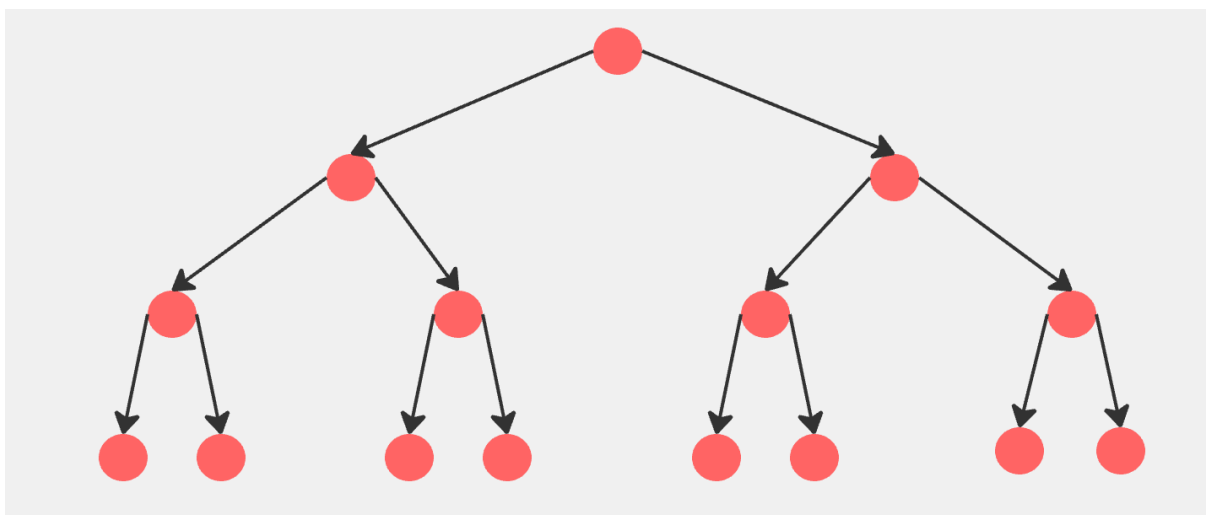


Image 5: network effect diagram example

In addition, after successful cases, we will ask the customer for permission to share them in sales meetings to help convince new customers to do business with us.

Two other strategies involve community outreach activities for awareness and brand positioning. However, for reasons of competitive strategy and know-how protection, these strategies cannot be disclosed at this time. Confidentiality is essential, as the company has not yet been launched on the market and premature exposure of these elements could compromise its competitive advantage at the time of its official introduction.

According to an article published by VOCÊ S/A in 2017, demand for digital investigators began to grow once companies started a process of “getting to know their employees” in order to understand their background before hiring them,

especially for management positions. With this in mind, the latest sales strategy will be launched once the portfolio of success stories is more advanced, and will aim to provide companies, especially banks (which carry out this process a lot), with a B2B service focused on background checks.

To maximize Customer Lifetime Value (LTV) and ensure business sustainability, the following retention strategies were created:

- **Delivery Excellence:** Ensuring that every intelligence report exceeds expectations is the foundation of our retention, turning one-time clients into brand promoters.
- **Recurring Monitoring:** Offering corporate and VIP clients "privacy maintenance" plans, with periodic scans (quarterly or semi-annually) to identify new data leaks, creating recurring revenue rather than just transactional income.
- **Post-Sale Relationship:** Maintaining an open communication channel for quick advisory, ensuring Unstaked is the first option consulted during any new crisis or security question.

2.5.4 Financial Projection and Feasibility:

Revenue Model & Pricing

The startup began with the idea of charging by the hour for research, but later, after simulations with a controlled group of people, this possibility proved not to be very viable, except for very complex cases. Therefore, the need for a future pivot was identified at the end of the project cycle. For the following analyses, the current definitions and projections were maintained as a basic feasibility indicator, serving as a solid parameter of technical effort versus return. The full transition of financial calculations to the new package model is therefore recognized as a necessary development for the next phase of implementation, ensuring that the final financial planning accurately reflects the commercial reality validated with the public.

Pricing Strategy

Initially, the technical hourly rate was calculated at R\$ 93.17 (based on the minimum operating cost of R\$ 32.94 + margin for ideal market salaries). This amount undergoes strategic adjustments to R\$ 100.00 in the third year and R\$ 105.00 in the fifth year.

For more details on the rationale behind the calculations, see APPENDIX D on the Appendices section.

Projected Expenses, Break-Even Point, and Feasibility Indicators

The cost structure was designed to be lean, maximizing profit margins as the company gains traction. Therefore, project expenses are:

- **Operating costs (OpEx):** include subscriptions to intelligence tools, which total approximately R\$ 2,086.00 per year at the beginning of implementation. However, as demand increases, and with it revenue, a larger amount will be allocated to subscriptions.
- **Infrastructure:** on-demand model using WeWork for face-to-face meetings (\$76/hour), eliminating fixed costs for idle office rent.
- **Human capital:** the opportunity cost of partners (estimated pro-labore): starts at R\$ 4,750.00/month and is projected to reach R\$ 9,750.00 in the 5th year, in line with seniority.

Considering these costs, the break-even point and feasibility of the project are then analyzed based on projections made for five years after launch:

- **Year 1 and 2 (Traction phase):** an operating loss is projected for the first year and the first half of the second year, which is common for service startups that depend on building reputation without massive investment in marketing.
- **Break-even point:** break-even is reached in September of the second year, when revenue exceeds costs, with earnings of R\$ 9,319.27 and costs of R\$ 8,881.74.

In addition, the project has two profitability indicators, the first being the fact that the profit margin tend to increase over the years as employees become more specialized, and the second considering the long term that the operation demonstrates high financial viability, with low fixed costs and high margin scalability (not volume), typical of specialized consulting firms.

To better understand the basis for cash flow calculations, see the APPENDIX E, on the Appendices section.

Initial Investment Requirement

To start the startup's operations with adequate technical capacity, the initial investment required is focused on training (knowledge) and software tools, since there are no expenses with inventory or heavy machinery.

Considering this, the initial composition of the investment is given by:

1. Technical training (knowledge setup)
 - Basic Course: R\$ 800.00 (one-time payment).
 - Advanced Course (International): R\$ 2,234.40 (annual).
2. **Software stack:** initial subscription to essential research tools costs around R\$ 170.00/month.

3. **Working capital:** strategic reserve to cover the operating deficit forecast for Year 1 and administrative costs (company start-up, accountant, and others).

Estimated Total: the initial direct investment (setup) is around R\$ 3,500.00 to R\$ 5,000.00, plus working capital to sustain operations for the first 12 months until break-even is reached.

2.6 Validation and Results

This section presents the practical validation process for the business model and technical feasibility of the solution. To ensure that the project goes beyond the theoretical field, the results obtained through direct contact with actual demand are detailed. In addition, key performance indicators (KPIs) are established that are fundamental for monitoring growth, concluding with a strategic analysis of the risks inherent in the operation and their respective mitigation plans to ensure business continuity.

2.6.1 Validation Methodology

Considering that the startup has not yet been launched on the market, we did not obtain a large amount of real, practical evidence that the solution and its business model actually work. Therefore, the way we found to validate the solution was to apply the processes defined in the IDEF0 diagram (section 2.4.3) in practice in a real use case, which has already provided many insights into how we should organize ourselves.

Section 2.4.4 provides further details on the investigative process and points to consider for future steps. In general, the structuring of the processes ensured the full execution of the stages. However, due to time constraints, the final delivery did not meet the standards of excellence established by the company, a point corroborated by customer feedback. Nevertheless, from a management and operational architecture perspective, it can be concluded that the internal processes are well defined and have demonstrated functionality and robustness.

In addition, simulations carried out with control groups allowed the viability of the original financial model to be tested, revealing that the hourly billing structure met with some resistance from consumers. Given this diagnosis, a strategic need was identified for a future pivot in the monetization model, pointing to a closed package structure (fixed price per investigation period) as a way to ensure greater predictability and accessibility for the customer. Under this new guideline, the hourly billing model would be reserved only for cases of exceptional complexity, aiming for greater alignment with the expectations gathered during market testing.

In this context, it is important to note that the way we earn money and the financial projections presented in this document were still consolidated based on the original hourly billing model. As the identification of this need for transition occurred at an advanced stage of the project cycle, it was decided to maintain the current definitions and projections as a baseline feasibility indicator, serving as a solid parameter of technical effort versus return. The full transition of financial calculations to the new package model is therefore recognized as a necessary development for the next phase of implementation, ensuring that the final financial planning accurately reflects the commercial reality validated with the public.

As well as the simulations and real-world use case, a small investigation was carried out with an influencer, which is described in detail in section 2.4.4. Without realizing the seriousness of the situation, this influencer posted several frames of the place where they live on their Instagram account. We were able to identify the location very easily, just by using image recognition. This example was used in a conversation with potential customers and conveyed a very positive image of the startup's work, as it proved very easy to reach someone who seemed unreachable just because of a video they posted.

2.6.2 Market Validation Results

As the startup has not yet been officially launched on the market, the data collected comes exclusively from internal observations, surveys with potential customers, and feedback provided by the customer in the case study.

First, a survey was conducted with potential customers to understand their views on this market. The form was sent to seven people, but only five responded, which was enough to gain many insights.

Most of the responses were from people over 35 and who hold leadership positions in the market. Of the five people interviewed, only three were aware of the digital investigation service, two through recommendations and one because of work issues, which reflects and proves that Brazilians still have little knowledge of this type of service. When they do find out about it, it's usually because of some significant personal need, rather than spontaneous curiosity or widespread dissemination.

Continuing with the answers, the majority said that they didn't know what information can be found through digital investigation, but that they understand the importance and need for the service. This factor was crucial, as it reflects that even though the service involves something very new, and people are not directly familiar with it, they understand its importance and how it can help them.

In addition, a relevant point observed was that all the respondents showed awareness of the leakage of their personal data on the internet and expressed fear that this information could be used as a form of threat against them. This shows that

the likelihood of us getting clients with the aim of knowing how exposed they are on the internet is high.

To finish, the majority reported that they value effective professionals, who are competent with a previous track record in other investigations and a good reputation, who show results and who are mature. This will certainly be a challenge in the beginning, as we will have few clients and low competence, considering that we are just starting out in the field. As reported in the risk mitigation strategies on section 2.6.4, to minimize this situation, we will focus on obtaining certifications that can prove our knowledge and always maintain a good relationship with the client for a good reputation.

Secondly, the feedback we received on the actual implementation of the case was that the final report presenting the data was too confusing to interpret, which had already been identified by the team shortly after delivery, since the short deadline and lack of evidence meant that the delivery of the final report was not given due attention, which will be a learning experience for future demands.

Finally, it is important to mention that at the end of the project, we had a moment of future pivoting based on feedback from simulations carried out with a control group. The point of discussion was whether charging the client per hour of investigation was the best way, since for some simpler cases, depending on the time it took, it could be much higher than expected. The simulations were carried out as if each individual in the group were a client, until we reached the conclusion that charges would be made in “packages” to make the price fairer. Each case has a price range, and only very large and complex cases will be charged by the hour.

In this package billing model, a deadline is set for the investigation, and the loss of taking too long on one case will be offset by those that are faster than expected.

In this regard, it is important to note that the need to make this transition occurred at an advanced stage of the project cycle, and therefore it was decided to maintain the hourly billing method and the financial projections presented previously as a basic feasibility indicator. The full transition of financial calculations to the new package model is therefore recognized as a necessary development for the next phase of implementation, ensuring that the final financial planning accurately reflects the commercial reality validated with the public.

2.6.3 Key Performance Indicators (KPIs)

To assess operational efficiency, runtime KPIs were created that will mainly measure how long it takes to conduct an investigation in each category. Although in the case of our service this time is very relative, taking these measurements will help us understand patterns and obtain an average, thereby adjusting both the amount to be charged to the customer in the packages and improving overall efficiency.

- Lead time of criminal investigations
- Lead time of threat assessment investigations
- Lead time of intelligence gathering investigations

It is important to note that within each category there are subcategories, such as in the case of intelligence gathering, which can be spouse, regional, or executive cases, and each of these will have its own Lead time KPI.

In addition, another important KPI that we will measure is **customer response time**. Our service is usually sought when the problem needs to be resolved urgently, so our response time must be short. For an initial response, the reply should be within 24 hours, and for further investigation, less than 12 hours.

Finally, for some sub-processes of the digital investigation as a whole, we defined the cycle time KPI. Its purpose is to analyze the complete duration of a process in order to identify bottlenecks. The sub-processes measured are:

- Cycle time between initial contact and first meeting
- Cycle time for opportunity analysis and proposal creation
- Cycle time for preparation of tools and work strategies
- Cycle time for internal evaluation of research quality

2.6.4 Risks and Mitigation Plan

It is an essential tool for identifying, assessing and mitigating the main risks associated with a company. As the area of digital investigation involves collecting and analyzing sensitive information using OSINT (Open Source Intelligence) techniques, it is essential to ensure that the process is carried out safely, ethically and in compliance with current regulations. However, there are many issues involving the customer's knowledge and experience of the service that need to be addressed to avoid complications and future problems.

In this analysis, risks are classified based on their likelihood of occurrence and the impact they could have on the business, providing a clear view of which threats or opportunities require priority attention. In addition, mitigating measures are proposed to minimize the negative effects of each risk and guarantee the continuity and reliability of the service.

Matrix

1. Low delivery of results due to being new to the area
2. Clients not trusting the brand because it is new to the market
3. Clients not trusting the brand because it avoids exposure on the internet

4. Not getting many clients as the service will be known by word of mouth
5. Receiving many clients through recommendations
6. Losing access to certain technologies used during research
7. Growing the number of clients, but not the workforce
8. People not seeing value in the service
9. Clients not accepting that sometimes the investigator may not find anything and assuming that the service was bad
10. Little acceptance of the service due to lack of knowledge of the area or difficulty in differentiating itself from competitors.
11. The credibility of the service depends on the client's opinion, and if it's negative we lose clients.
12. The credibility of the service depends on the customer's opinion, and if it is positive we gain clients
13. Use of data obtained via OSINT can raise questions about privacy and legal compliance (LGPD, GDPR, etc.).

Probability		Threats					Opportunities				
Very High	5				2/3	10					
High	4			4		11	12				
Medium	3				1	8/9	5		7		
Low	2				6						
Very Low	1				13						
		1	2	3	4	5	5	4	3	2	1
		Very Low	Low	Medium	High	Very High	Very High	High	Medium	Low	Very Low
		Impact									

Table 2: Risk Matrix

Mitigations Strategies

For risks 1, 9 and 11, which relate to the customer's opinion of the service and how this reflects on our service in a mostly negative way, it was decided to adopt a mitigation strategy of reduction.

The reason is that we can't mold people to our will. As much as we strive to avoid negative opinions, they don't depend on us alone. Therefore, reducing these comments will already be a big step forward.

In this case, I want to make every specification very clear in the contract so that there are no future frustrations or broken expectations. In this process, it will be very important to emphasize to the client that not finding the information requested, or finding just a little, does not mean that we did a bad job, since there are situations in which the focus of the investigation does not have as much exposure on the internet.

To mitigate risks 4, 8 and 10, which are related to the difficulty of accepting the service in the market due to the lack of visibility and the clients' lack of knowledge about the area, a combined acceptance and prevention strategy was adopted.

We recognize that it's not possible to make people aware of the service instantly, especially considering that we won't be using social media to promote it. However, our website will have a section dedicated to explaining what digital investigation is, how it works and in what situations it can be useful. In this way, clients will be able to become more familiar with the concept, understanding its value and assessing whether it really is the solution they are looking for.

For risks 2 and 3, which focus on not contracting the service because it is a very new company, we will use as a mitigation strategy adding certifications and testimonials of success that we have obtained along the way to the website, so that we can pass on more credibility to the client with regard to the service, even though it is a new company on the market.

To mitigate risk 6, the main strategy is to increase the level of professionals' knowledge. Although some tools make it easier and faster to obtain information, they are not the only way to access data. With the right knowledge, it is possible to achieve the same results using different methods, thus guaranteeing the effectiveness of the service even in the face of technological limitations.

Risk 7 does not have a mitigation strategy, as this depends very much on the current situation the company finds itself in. Sometimes it's better to continue with a small workforce and serve fewer clients than to hire someone to grow the team.

Finally, for risk 13, it will also be specified in the contract signed by the client that all information is obtained ethically and legally, so that if we are questioned about the source of this information, it will be possible to prove it without causing damage to either side.

3 Conclusion

After the project was completed, it can be concluded that the main objective was achieved, since its focus was to develop the idea of a startup that offers digital investigation services, preparing all the necessary processes for its launch on the market in the first half of 2026.

The main requirement at the outset was to understand the digital investigation market and analyze competitors to determine what to offer that was different. The mapping was very thorough, and it enabled us to identify where and how to stand out.

The project continued with the creation of the landing page, which managed to convey the company's personality and make it clear to the client why digital

investigation is important. In addition to this, the process was detailed using the IDEF0 diagram, which allowed for a much more complete visualization of all the startup's needs at all stages of the service, supported by the tools used and techniques applied.

One of the most important steps involved understanding the project costs in order to define the billing method and price for the client. With these values mapped out, we made financial projections for the next 5 years, which allowed us to understand issues such as the break-even point.

The last stage of development focused on preparing launch strategies, creating an execution schedule for marketing strategies, and defining how customer onboarding will work once demand arrives. These definitions were important for assessing the feasibility of each strategy and preparing us early on for the arrival of customers.

Although we achieved most of the main objective, some points will remain pending as future actions for the startup to be launched correctly, which are:

- Talk to direct competitors to understand how the market is doing, which services are in highest demand, and what billing model they use;
- Pivot the current billing model to a package model, updating financial projections and making the service more accessible to customers;
- Launch the project in the first half of 2026;
- Launch a major marketing campaign to gain recognition purely through organic traffic and start operations;

Final thoughts: overall, the project achieved its main goal, and we learned a lot during development, which increased the company's maturity.

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Appendices

APPENDIX A - Table containing competitors analysis.

APPENDIX B - IDEF0 Diagram.

APPENDIX C - Business Model Canva.

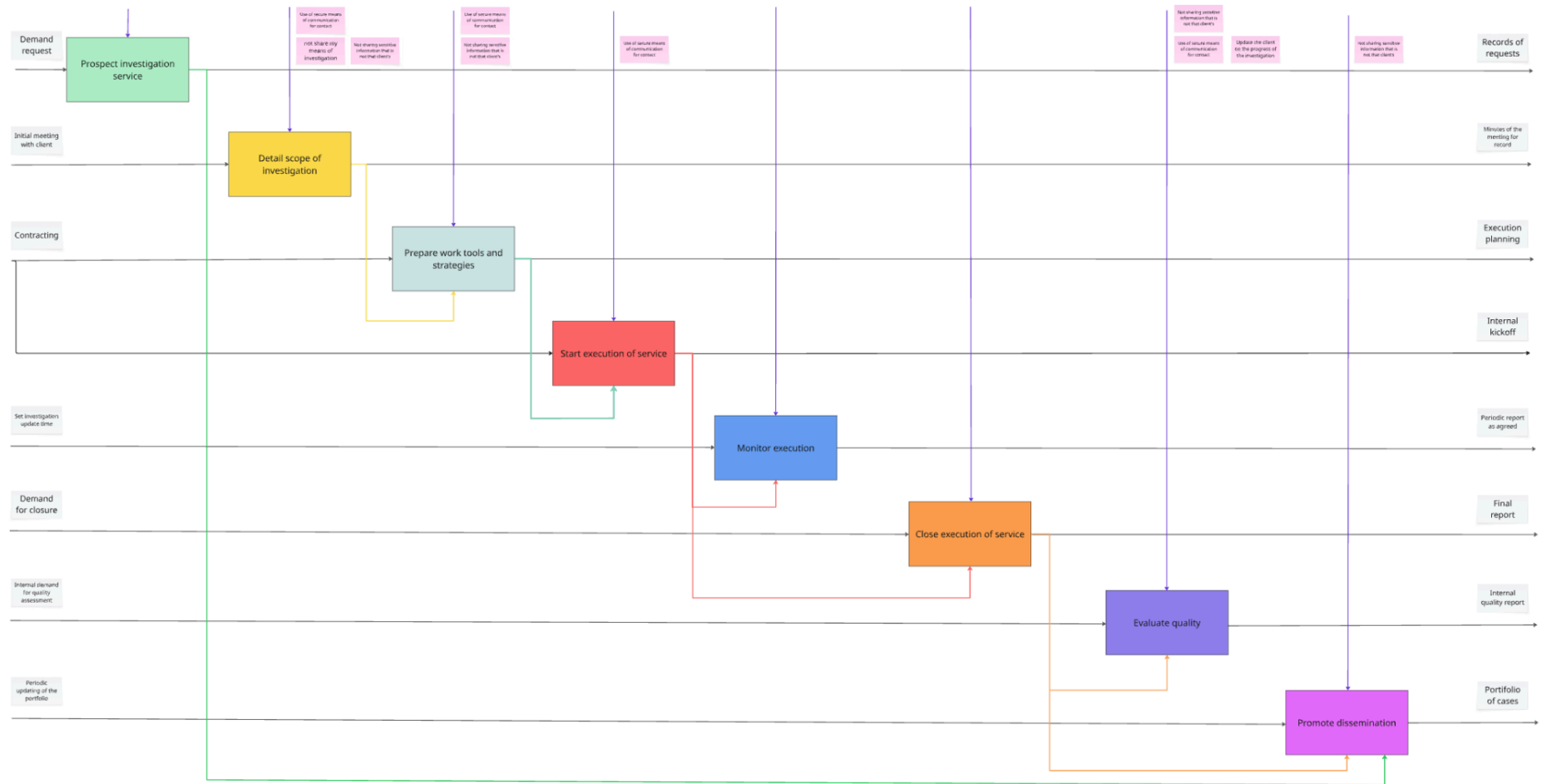
APPENDIX D - Costs assumptions report.

APPENDIX E - Discounted cash flow projection for 5 years.

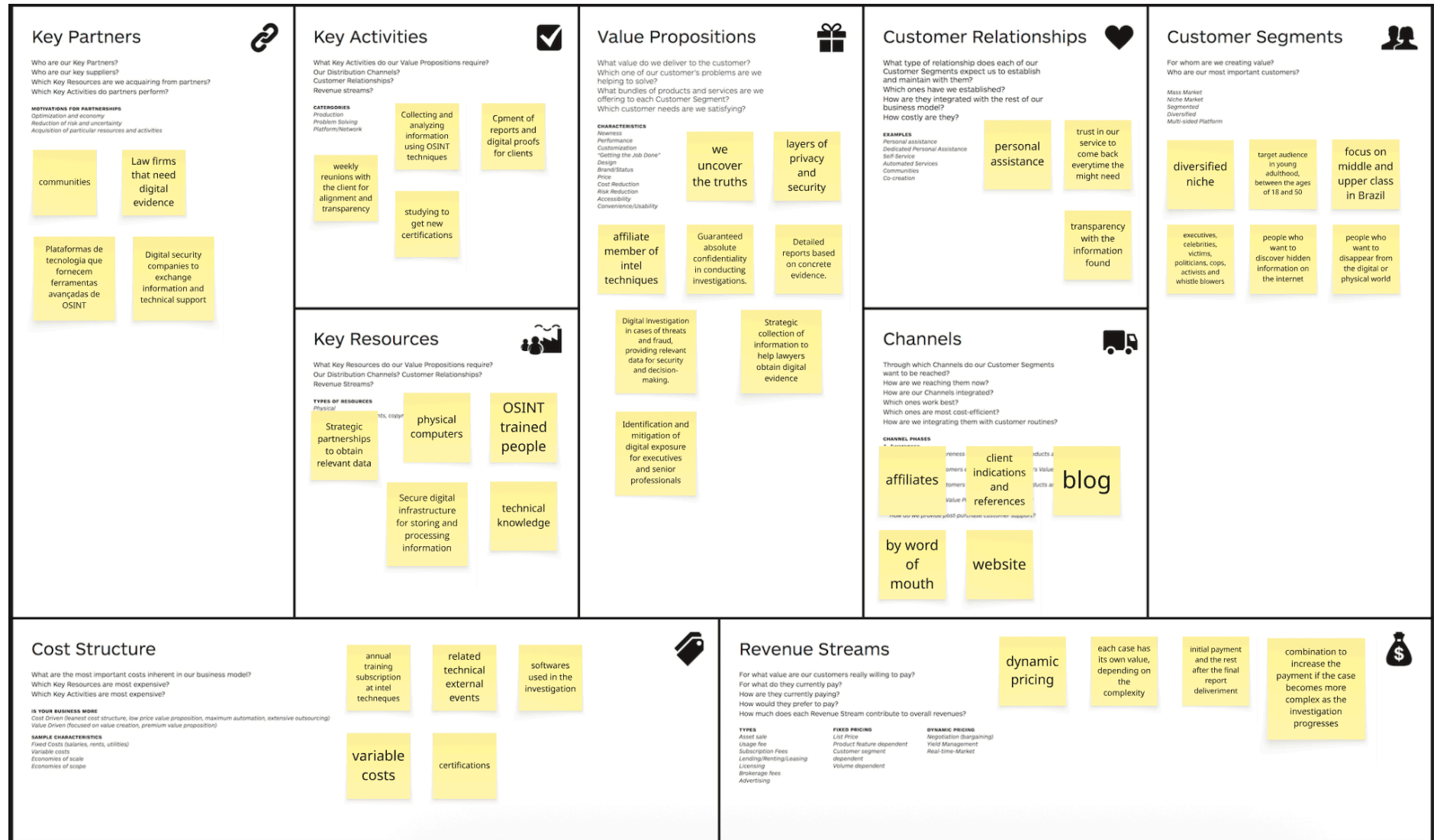
APPENDIX A - Table containing competitors analysis

Nome	O que oferece?	tipo de concorrente	Tempo no mercado	Pontos fortes	Pontos fracos	site
Bruno Fragax	It offers digital investigation services, mainly OSINT of all kinds, both low complexity, such as finding addresses, telephone numbers, etc. of an individual, and high complexity, such as making connections between people, finding criminals, etc.	direct	doesn't mention	- He was one of the pioneers in the digital research market; - He has created several courses on the subject, which shows his mastery; - He is well known in this market because he has been involved in cases involving famous people, which increases confidence in his service;	- It has built up a negative image over time with its course platform, because it promised and didn't deliver, as well as many other issues that can lead customers to no longer trust its services; - It no longer has a website or any other means of communicating its service, but only a YouTube channel, which it publishes weekly;	It doesn't have https://mirandadetective.com.br/sita/
Antônio Miranda	He offers a detective service, i.e. he uses the physical world to find out information, whether for family, business or private cases, among others. He carries out infiltrations if necessary, but always guarantees that the contractor will not be exposed and will have maximum security. His aim is to help the contractor obtain evidence to be used in court.	indirect	over 20 years' experience in private investigation	- Many years of experience in the market, which gives clients confidence; - Organized website with all the services it provides, covering various possibilities;	- Marketing for hiring a bit forced, - The links to your social networks on the website don't work;	https://detectivedessociais.com.br/
Detetive conjugal internacional	Promises to find out everything the contractor wants to know about their spouse	indirect	doesn't mention	- They serve several countries	- Very small site with very little information, which does not give confidence in the service; - From the structure of the site, it looks like a scam	https://www.detetiverodrigues.com.br/index.html
Agência detetive Rodrigues	Because it is an agency and therefore has more manpower. Detective Rodrigues covers the following services: marital investigations, family investigations, abuse investigations, business investigations, counter-espionage investigations and the sale and rental of devices. It is important to note that all investigations are carried out in the field.	indirect	over 20 years' experience in investigations	- Very well-structured website with several explanations and demonstrations of the service, which gives customers confidence; - She has a blog, where she publishes things related to her service, which complements the information on her website and shows that she has mastery of the subject; - has very well-defined services, and puts a lot of emphasis on marital cases, as these are the ones that attract the most clients - He makes it clear why the client should hire his service - Certified by the National Association of Detectives and Private Investigators of Brazil	- The About Us page doesn't work, but this hasn't harmed the company's credibility too much, as it goes over this point lightly on the home page	https://cyberexperts.com.br/
Cyber Experts	They offer services in digital and computer forensics, data protection and auditing, online reputation and privacy, forensics and cryptocurrency tracking	direct	more than 20 years of legal expertise and technical assistance in computers and information technology	- A pioneer in digital forensics - Recommended by JusBrasil in an article on digital forensics - They have an academy area with courses and training - They have a blog, which publishes things related to their service, which complements the information on their website and shows that they have mastery of the subject; - They have a page dedicated to explaining what digital forensics is, as it is not very well known; - They are serious in their words and in their presentation of the company, which gives the client more confidence; - They use the name digital forensics instead of digital investigation, which makes the service sound more serious;	- Site with a lot of text, which makes reading tiring and doesn't highlight the most important information	https://sabrainteligencia.com.br/servicos/pericia-forense-digital/
Sabra inteligência e investigação	It has a strong focus on corporate investigations, offering a range of services in this area, such as fraud, employee, executive, supplier, asset and background check investigations. It also offers computer forensics and cell phone forensics. In general, the service this company provides is related to uncovering crimes and fraud within the corporate world.	direct	more than 20 years	- Present the company's values and mission	- The visual identity of the site is not in keeping with the service, as the colors and images used are very reminiscent of a corporate health service, rather than a research service - The information provided on the site is not well presented, with a relative amount of text that is very similar - The blog page has no content, and once accessed it disfigures the entire site	https://stwtbrasil.com/
STW Brasil	They offer digital forensics services, which are more investigative, educational, offering training in the area through the Digital Forensics Academy, and auditing, consultancy, cybersecurity and support, which are more focused on the business sector.	direct	20 years of excellence in technology and security	- They have a blog, where they publish things related to their service, which complements the information on their website and shows that they have mastery of the subject; - They use the name digital forensics instead of digital investigation, which makes the service sound more serious; - They have several certifications that prove the quality of the service and the professionals - They have an educational area (Digital Forensics Academy), specialized in training individuals who want to work in the area - They have a space on their website dedicated to showing clients who have already hired the company and institutions trained by their academy - They have a podcast channel - They use the name digital forensics instead of digital investigation, which makes the service sound more serious;	- It exposes 4 of the professionals who work in the company with a high position, and since it is a company that provides security services, it is not recommended, since the data is exposed and this can damage the company's image in the eyes of the client.	

APPENDIX B - IDEF0 Diagram.



APPENDIX C - Business Model Canva.



APPENDIX D - Costs assumptions report.

Introduction

When providing services, it is very important to know the company's costs in order to calculate each employee's hourly rate. Therefore, this document focuses on the company's main current costs, considering future costs in case demand for the service increases.

In addition to providing a clear overview of the costs involved, this report also serves as a basis for supporting the company's strategic planning. With the identification and detailed analysis of costs, it becomes possible to assess the feasibility of new projects, set more competitive prices, and ensure long-term financial sustainability.

Costs

As this is a service provision, the costs are concentrated on training the two partners to improve their knowledge in the area, on the tools used during the investigation, and on the hourly cost of the work itself.

Some of the values shown below are in US dollars, and considering their daily variation, the calculation was made using an exchange rate of **5.6 Brazilian reais to 1 US dollar**. This ensures a small margin of safety in case the value increases during the execution of a project.

Courses

To assist in training and improving digital investigation skills, two courses were purchased:

- **Basic Course:** course in Portuguese that introduces the basic techniques used when conducting an investigation.
 - The **total amount** was **R\$ 800.00**, a one-time payment for lifetime access.
- **Advanced Course:** English course provided by a well-known professional in the field. This course is updated every year and is therefore more detailed than the first one.
 - The **annual cost** is **\$399**, which converts to **R\$ 2,234.40**.

As the Basic course is not a recurring cost, it will not be considered as a recurring cost. IntelTechniques, on the other hand, is an annual cost, so the amount will be divided into 12 monthly payments. Furthermore, it is important to note that as the course is priced in US dollars, its value may vary.

Tools

Currently, as we do not have recurring demand, we only subscribe to one monthly tool, at a cost of **R\$ 45.00 per month**. However, if demand increases, it will be necessary to use more tools, and their costs are outlined below, however, some names are masked so as not to compromise the integrity of the service:

- **Tool one:** the **monthly cost** is **\$21.99**, which converts to **R\$ 123,14**.

- **Hunch.ly:** software that assists in the process and organization of research, creating connections between the data found and making it easier to document the information. The **annual cost** is **\$109**, which converts to **R\$ 610,40**.
- **Tool 2:** the **annual cost** is **€2.500**, which converts to **R\$ 15.871,00**.

Furthermore, considering that we are beginners in this field and will be constantly seeking to improve our knowledge, it is very likely that new tools will emerge along the way, but for now, the ones mentioned above are already of great help.

Certifications

To prove our knowledge and bring more visibility to our service, it is highly recommended that my partner and I obtain certifications in the field. For now, the goal is to focus on obtaining OSIP certification, which is provided by IntelTechniques at a cost of **\$300**, which converts to **R\$ 1.680,00**.

This cost will not be included for the time being, as much research is still needed to achieve this goal.

Opportunity costs

To determine whether a venture will be able to replace your current source of income, your salary and other sources of income are considered as costs. In this case, we will consider the amount of **R\$ 4,750**. This amount represents the average between the ideal salary for a junior and a pleno. The reason why the account was set up this way is that the company currently has only two partners, myself as a junior partner and the other as a pleno partner.

Infrastructure

The best option in this case is to use a coworking space. This is because, as there are very few employees, there is no need to work in an office and bear the costs of one.

Even so, a coworking space is necessary for face-to-face meetings, both at the beginning of the contract and at the end for the delivery of results.

In a conversation with the owner of Gombo, one of the leading coworking agencies on LinkedIn, we got some good guidance on which agency to use and the costs associated with this option. He recommended using WeWork, which is well suited to many different needs, especially on demand (which is our case).

The company offers several meeting rooms, and the larger the room, the higher the hourly cost. We will use rooms for four people, which cost **R\$ 76,00 per hour**.

Taxes

Here, it is not necessary to provide many details about figures, as this is more of a private matter for the company. However, in a conversation with the accountant, it was mentioned that the tax rate on total revenue is **6%**.

Other costs

On a monthly basis, to organize the company's finances, we will need an accountant, who will do what is necessary at the end of the month for a fee of **R\$ 300.00**, since we do not yet have a high turnover, meaning that the amount may increase in the future.

In addition, it is normal for accountants to receive a thirteenth month's salary, so the calculation was made considering 13 months.

Conclusion

After performing all conversions and arriving at an hourly value for all current costs, it was possible to understand the minimum we should charge per hour, considering a salary equivalent to the opportunity cost. In this scenario, our **hour is worth R\$32.94**, but it is important to note that the tax amount has not yet been applied.

APPENDIX E - Discounted cash flow projection for 5 years.

Introduction

Discounted cash flow is an essential tool for assessing the economic value of a company or project, projecting its ability to generate financial results over time. This methodology considers expected future cash flows and brings them to present value by applying a discount rate that reflects the risk and opportunity cost of capital. This provides a more accurate and realistic view of the intrinsic value of the business.

This document presents a discounted cash flow analysis with a 5-year projection, allowing for an assessment of both the growth potential and the risks associated with the operation. The objective is to provide a consistent overview to support strategic decisions, balancing revenue, cost, and investment expectations in order to support the company's long-term sustainability and value appreciation.

Costs projections

Over the five years, some costs have changed, and others have been added as needed due to the company's growth, including: amount reserved for new tools and certifications, WeWork adjustments due to increased demand, and changes in opportunity cost due to employee knowledge gains.

Tools

In general, each year saw the acquisition of three or four new tools. In 2026, the first two have already been defined, while the others will be discovered over time.

The first tool is acquired in April 2026 at a cost of R\$123.14 per month. Then, in June, Hunchly was acquired, which will help organize investigation information at a monthly cost of R\$610.00. Also in June, an amount of R\$150.00 was set aside for the acquisition of another tool.

After 2026, there were no new tool purchases. From 2027 onwards, as no new items were defined, the reference value of R\$150.00 (as mentioned above) was adopted as the basis for calculations. This amount represents an average, considering that many tools have a monthly cost lower than this amount, while others may be more expensive. The intention is that, at the end of each year, the estimated amount will be sufficient to cover variations, even if it does not correspond exactly to the monthly expenditure in all periods.

Wework adjustments

As demand increases, more meetings will be necessary, and therefore the cost of using WeWork will increase proportionally in line with growth. The year 2026 begins with a monthly fee of R\$304.00, which covers four one-hour meetings. This amount was designed for two clients, with one meeting for briefing and another for delivering results. Changes may be made if a client also wants face-to-face meetings throughout the investigation or if they do not see the need to travel to share information.

Over the five-year projection period, demand growth was not uniform: in 2026, there were two direct jumps of 10%, while from 2027 onwards, the variation was fixed at 5% in hours worked. From this, a rule of three was applied to calculate the adjustment to the cost of WeWork. As the starting point was 23.4 hours per month and R\$304.00 for WeWork (equivalent to 100%), it was sufficient to adjust the proportions to find the appropriate value, which resulted in R\$152.00. In practice, this calculation suggests that each 5% increase in demand would correspond to one new client. However, as the time dedicated can vary between projects, this relationship should not be interpreted as an absolute rule.

At the end of the five years, the monthly cost of WeWork, considering an average of 235.2 hours (assuming that employees will work individually), will be R\$3,052.15.

Certifications

To bring more credibility to our service, we have set aside an amount for two important certifications. The first will take place in July 2027, at a cost of R\$3,360.00 for two employees, and will be provided by IntelTechniques. The second certification has not yet been defined, but will be carried out around June 2029, and an amount of R\$2,000.00 has been set aside for each employee (totaling R\$4,000.00). This amount is based on the average cost of certifications in the area.

Change on opportunity costs

In the first year, the opportunity cost is R\$4,750.00 (average between the Junior and Pleno salary). Over the course of five years, it was necessary to adjust this amount to remain consistent with the reality and professional development of employees. As such, there were three updates.

The first change occurred in January 2027, with a salary adjustment. The junior employee, who previously earned R\$3,500.00, now earns R\$4,500.00, while the pleno had their salary adjusted from R\$6,000.00 to R\$7,500.00. This increase generated an increase in opportunity cost of R\$1,250.00, bringing the total to **R\$6,000.00**.

The second change occurred in January 2029, with a promotion. Although the evolution of professionals occurs at different times, for estimation purposes, both were promoted at the same time. The junior employee became a pleno employee with a salary of R\$6,000.00, and the pleno employee became a senior employee with a salary of R\$10,000.00. The average opportunity cost totals **R\$8,000.00**, representing an increase of R\$2,000.00 compared to the last change.

Finally, in June 2030, due to increased demand, a new salary adjustment was made. Junior employees, who previously earned R\$6,000.00, now earn R\$7,500.00, while senior employees had their salaries raised from R\$10,000.00 to R\$12,000.00. As a result, the total salary cost rose to R\$19,500.00, reflecting an opportunity cost of **R\$9,750.00**, only R\$1,750.00 above the value of the last change.